Lecture 8

Tangent spaces, Lie groups and homogeneous spaces

Objectives. *A tangent vector records the initial velocity of a curve without choosing an ambient Euclidean space. Coordinate changes relate its components by their Jacobian matrices. We use these vectors to differentiate maps and describe the infinitesimal directions of matrix groups. A transitive smooth action of a Lie group presents a manifold as a quotient by the subgroup fixing a point. The sphere and the qubit state space give two concrete quotient descriptions.*

By the end of this lecture you will be able to:

- Define tangent vectors by curves and transform their components.
- Compute differentials and tangent spaces to regular level sets.
- Identify the tangent space of $\mathrm{SO}(3)$ at its identity.
- Determine stabilisers and justify the sphere quotient descriptions.

Unit 8A Tangent spaces

8.1 Tangent vectors as velocities

Let p be a point of a smooth n-manifold M.

Definition 8.1 (Tangent vector)**.** A *smooth curve through* p is a smooth map $\gamma\colon (-\varepsilon,\varepsilon) \to M$ with $\gamma(0) = p$, where $\varepsilon > 0$. In a chart $x = (x^1,\dots,x^n)$ near p, its coordinate velocity is $(x\circ\gamma)'(0) \in \mathbb{R}^n$. Two such curves are equivalent if these velocities agree. A *tangent vector* at p is an equivalence class of curves.

Definition 8.2 (Tangent space)**.** The set of these classes is the *tangent space*, written T_pM.

The drawing shows curves with different second-order behaviour but the same initial velocity in one chart.

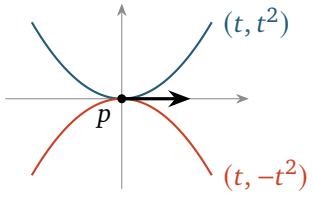


Figure 8.1: The coordinate curves (t, t^2) and $(t, -t^2)$ have the same velocity $(1, 0)$ at the marked point. Their common tangent vector does not record their bending.

Theorem 8.3 (Velocity classes and components). *Curve equivalence is independent of the chart. A chart identifies T_pM with $\mathbb{R}^n$, giving it an n-dimensional vector-space structure independent of that chart. Under a coordinate change $y = y(x)$, components transform by*

$$v_y^a = \frac{\partial y^a}{\partial x^i}(p) v_x^i.$$

Proof. The chain rule gives $(y \circ \gamma)'(0) = D(y \circ x^{-1})_{x(p)}(x \circ \gamma)'(0)$. Thus equal velocities in x give equal velocities in y. The reverse coordinate change proves the converse. This also gives the displayed component formula.

The velocity map $T_pM \to \mathbb{R}^n$ is injective by the definition of equivalence. For any $v \in \mathbb{R}^n$, the curve $\gamma(t) = x^{-1}(x(p) + tv)$ is defined for sufficiently small $|t|$ because the chart image is open. It has velocity v, proving surjectivity. Transport addition and scalar multiplication through this bijection. A change of chart acts by an invertible linear Jacobian, so it preserves these operations. This proves that the vector-space structure is independent of the chart. ∎

The basis vector with coordinate components equal to the ith unit column is written $\partial_i|_p = \partial/\partial x^i|_p$. Thus $v = v^i \partial_i|_p$, with the upper/lower index convention of Lecture 7. The symbols $\partial_i|_p$ denote tangent vectors, not points of M.

A tangent vector differentiates smooth functions near p by

$$v(f) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} f(\gamma(t)) = v^i \frac{\partial (f \circ x^{-1})}{\partial x^i}(x(p)).$$

The right side shows independence of the representative curve.

Definition 8.4 (Point derivation). A *point derivation* at p is a real linear rule on smooth real functions near p satisfying the product rule

$$v(fg) = v(f)g(p) + f(p)v(g).$$

The curve formula is real linear in the function. The ordinary product rule therefore makes every tangent vector a point derivation. Applied to a function, the vector $\partial_i|_p$ gives its partial derivative in the chart. Lecture 9 uses this action to differentiate functions by vector fields.

We obtain the change of coordinate basis by applying the component law to each basis vector. Its x-components form the ith unit column, so

$$\left.\frac{\partial}{\partial x^i}\right|_p = \frac{\partial y^a}{\partial x^i}(p) \left.\frac{\partial}{\partial y^a}\right|_p.$$

The vector stays fixed while its basis and its component column change. The figure shows one vector in Cartesian and polar bases at the same point.

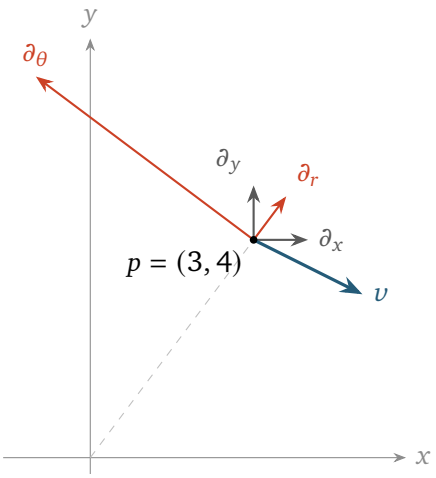


Figure 8.2: The vector with Cartesian components $(2, -1)$ at $p = (3, 4)$ has polar components $(2/5, -11/25)$. The polar angular basis vector has length 5, so it is not a unit vector.

Example 8.5 (One vector in two charts)**.** We express the velocity of a curve in Cartesian and polar coordinates. The point and curve are

$$p = (3, 4), \qquad \gamma(t) = (3 + 2t, 4 - t).$$

We use a local polar chart with $r > 0$ and a smooth angle branch near $\theta = \arctan(4/3)$. The Cartesian velocity column is $(2, -1)^T$. The Jacobian rows of $r = \sqrt{x^2 + y^2}$ and of the local angle give

$$\frac{\partial(r, \theta)}{\partial(x, y)} = \begin{pmatrix} x/r & y/r \\ -y/r^2 & x/r^2 \end{pmatrix}, \qquad \left.\frac{\partial(r, \theta)}{\partial(x, y)}\right|_p = \begin{pmatrix} 3/5 & 4/5 \\ -4/25 & 3/25 \end{pmatrix}.$$

The component law now gives

$$\begin{pmatrix} v^r \\ v^\theta \end{pmatrix} = \begin{pmatrix} 3/5 & 4/5 \\ -4/25 & 3/25 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 2/5 \\ -11/25 \end{pmatrix}.$$

For the basis conversion we use $x = r\cos\theta$ and $y = r\sin\theta$. The columns of this inverse Jacobian give

$$\partial_r|_p = \frac{3}{5}\partial_x|_p + \frac{4}{5}\partial_y|_p, \qquad \partial_\theta|_p = -4\partial_x|_p + 3\partial_y|_p.$$

Substitution recovers the original vector:

$$v = \frac{2}{5}\partial_r|_p - \frac{11}{25}\partial_\theta|_p = 2\partial_x|_p - \partial_y|_p.$$

The two columns describe the same velocity class.

8.2 Germs and the converse derivation characterisation *(optional)*

The differentials, vector fields and group quotients in later lectures do not depend on germs or on the converse characterisation in this section.

Definition 8.6 (Germ)**.** A *germ* at p is an equivalence class of functions defined near p, with two functions equivalent when they agree on some smaller neighbourhood.

Only the function near p matters in the action of a tangent vector. The product rule therefore makes sense on germs. Conversely, all point derivations on smooth germs have the displayed coordinate form for $v(f)$ in §8.1. This comparison is cited from Tu, §8, Proposition 8.9.

8.3 Differentials and regular constraints

Definition 8.7 (Differential). For a smooth map $f\colon M \to N$, its *differential* at p is

$$\mathrm{d}f_p\colon T_pM \longrightarrow T_{f(p)}N, \qquad \mathrm{d}f_p[\gamma] = [f\circ\gamma].$$

In coordinates the image velocity is $D\widehat{f}_{x(p)}v_x$, by the chain rule. This depends only on the velocity class and is linear. It therefore defines the stated linear map. For $g\colon N \to P$, composition on representatives gives the intrinsic chain rule

$$\mathrm{d}(g\circ f)_p = \mathrm{d}g_{f(p)} \circ \mathrm{d}f_p. \tag{8.1}$$

The identity map has identity differential. Applying this formula to a diffeomorphism and its inverse shows that their differentials are inverse linear maps.

For example, if $f\colon \mathbb{R}^2 \to \mathbb{R}^3$ is given by $f(s,t) = (s,t,s^2+st)$, then

$$\mathrm{d}f_{(s,t)}(u,v) = (u,v,(2s+t)u+sv).$$

Its first two components recover (u,v), so the differential is injective.

Definition 8.8 (Immersion, submersion and embedding). Recall from Lecture 7 that an injective differential defines an *immersion*. An onto differential defines a *submersion*. An immersion that is a homeomorphism onto its image is an *embedding*.

Proposition 8.9 (Tangent space of a level set · *proved given the implicit function theorem*). *Let $S = f^{-1}(c) \subset W \subseteq \mathbb{R}^N$ be a nonempty regular level set of a smooth map $f\colon W \to \mathbb{R}^k$, with W open. The inclusion identifies T_pS with $\ker Df_p$ for every $p \in S$.*

Proof. For a curve in S, differentiating $f(\gamma(t)) = c$ gives $Df_p\gamma'(0) = 0$. In the graph chart of Theorem 7.19, the inclusion has the form $s \mapsto (s, h(s))$. Its differential sends u to $(u, Dh\,u)$, so it is injective. Its image has dimension $N-k$. The kernel of the onto map Df_p has that same dimension by rank–nullity. The inclusion already proved is therefore equality. ∎

For S^2, the constraint $|p|^2 = 1$ gives $T_pS^2 = \{v \in \mathbb{R}^3 : p\cdot v = 0\}$, since the derivative is $v \mapsto 2p\cdot v$. Thus the intrinsic velocity classes agree with the familiar tangent plane in this embedded example. The figure places velocity arrows in the affine planes through their base points.

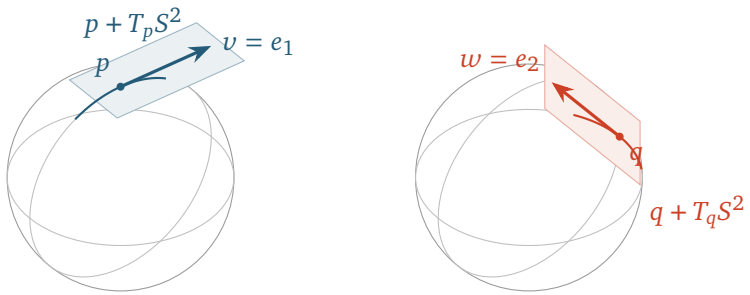


Figure 8.3: The planes $p + T_pS^2$ and $q + T_qS^2$ touch the sphere at $p = e_3$ and $q = e_1$. Each great circle has its velocity arrow in the plane at its own base point.

The vector spaces T_pS^2 and T_qS^2 are different subspaces of $\mathbb{R}^3$. Their affine translates in the figure show where their vectors are based. On a general manifold there need not be a common ambient vector space.

8.4 The tangent bundle

A moving point has both a position and a velocity at that position. We collect these pairs in one space so that the velocity can vary along a curve.

Definition 8.10 (Tangent bundle)**.** The *tangent bundle* is the disjoint union of all tangent spaces:

$$TM = \{(p, v) : p \in M,\ v \in T_pM\}.$$

The projection $\pi_{TM} \colon TM \to M$ is given by $\pi_{TM}(p, v) = p$. Its *fibre* over p is the inverse image $\pi_{TM}^{-1}(p)$, identified with T_pM.

Vectors at different base points belong to different fibres. They cannot be added without a further identification.

Physical input. In a mechanical model with configuration manifold M, the instantaneous position and velocity form a pair (p, v) with $v \in T_pM$.

Thus a time-independent Lagrangian, which assigns a real number to each position and velocity, is a function $L \colon TM \to \mathbb{R}$. For a pendulum the configuration is a point of S^1. Its position–velocity states therefore form TS^1, the cylinder drawn below.

Proposition 8.11 (Manifold structure on the tangent bundle · *sketched*)**.** *If M is a smooth n-manifold, then TM has a natural smooth structure of dimension $2n$. The projection is a smooth submersion.*

We sketch its chart construction. A chart x on $U \subseteq M$ gives coordinates $(x(p), v_x)$ on $\pi_{TM}^{-1}(U)$ with image $x(U) \times \mathbb{R}^n$. If y is another chart, the overlap map is

$$(x, v_x) \longmapsto \big(y(x), Dy_x v_x\big).$$

It is smooth and has the inverse obtained by the inverse coordinate change. The base and velocity coordinates give $2n$ independent real coordinates. They also specify the topology. Distinct base points can be separated downstairs. Distinct vectors over the same point can be separated in a common bundle chart. A countable atlas downstairs gives a countable basis in these product coordinates. This gives Hausdorffness and second countability. In each chart the projection is $(x, v) \mapsto x$, whose derivative is onto. On a triple overlap, the

ordinary chain rule makes these transition maps compose correctly. The inverse transition returns the original velocity because the two coordinate Jacobians are inverse matrices. These checks give compatible bundle charts. Further details are in Tu, §12.

For the circle, write $p = (\cos\theta, \sin\theta)$. Every tangent vector has the unique form $v = r(-\sin\theta, \cos\theta)$ with $r \in \mathbb{R}$. These formulas identify TS^1 with $S^1 \times \mathbb{R}$: the angle is periodic but the velocity coefficient is unrestricted. The figure displays this cylinder and its projection.

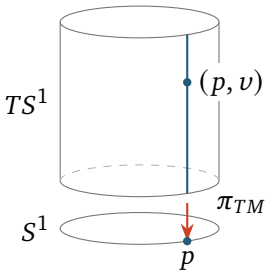


Figure 8.4: A fibre of TS^1 is a line of velocity coefficients. Projection forgets the coefficient and retains the point of the circle.

An identification with $M \times \mathbb{R}^n$ that preserves the projection and is linear on each fibre would choose a basis smoothly in every tangent space. The tangent bundle of S^2 admits no such identification. Indeed, the first basis vector would give a smooth tangent field with no zero. Lecture 14 proves that no such field exists.

Self-check. At $p = (-4, 3)$, let $v = 2\partial_x|_p - 3\partial_y|_p$. Use a local polar chart with angle near the second quadrant to find its polar components and recover v from the polar basis. For the map $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ given by $f(x, y) = (x^2 - y, xy)$, compute $\mathrm{d}f_p(v)$ and state its base point.

Answer. The two Jacobian rows at p are $(-4/5, 3/5)$ and $(-3/25, -4/25)$. Multiplying them by $(2, -3)^T$ gives $(-17/5, 6/25)^T$. The inverse basis vectors are

$$\partial_r|_p = -\frac{4}{5}\partial_x|_p + \frac{3}{5}\partial_y|_p, \qquad \partial_\theta|_p = -3\partial_x|_p - 4\partial_y|_p.$$

Their linear combination with these polar components is v. For the differential we obtain

$$\mathrm{d}f_p(v) = \begin{pmatrix} -8 & -1 \\ 3 & -4 \end{pmatrix} \begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} -13 \\ 18 \end{pmatrix} \quad \text{at } f(p) = (13, -12).$$

Self-check. State in one sentence what a tangent vector is and why we introduced it. Then describe a point of TS^1 and what its projection retains.

Answer. A tangent vector is a class of curves through one point with the same coordinate velocity, introduced to define velocity without an ambient space. The chain rule makes equality of those velocities independent of the chart. A point of TS^1 is a circle position together with one tangent velocity there. The projection retains the position.

The tangent space at the identity of a smooth group describes the velocities of curves of symmetries that start at the identity.

Unit 8B Lie groups and homogeneous spaces

8.5 Lie groups and tangent matrices

Definition 8.12 (Lie group)**.** A *Lie group* is a smooth manifold G that is also a group, with smooth multiplication $G \times G \to G$ and smooth inversion $G \to G$. Write e for its identity.

Definition 8.13 (General linear group)**.** For $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, the *general linear group* is

$$\mathrm{GL}(n, \mathbb{K}) = \{A \in \mathbb{K}^{n \times n} : \det A \neq 0\}.$$

Definition 8.14 (Special orthogonal group)**.** The *special orthogonal group* is

$$\mathrm{SO}(n) = \{A \in \mathrm{O}(n) : \det A = 1\}.$$

Definition 8.15 (Matrix subgroup)**.** A *matrix subgroup* is a subgroup of a general linear group.

For complex matrices, the ambient space is regarded as a real manifold.

Proposition 8.16 (Regular matrix groups · *proved given the regular level set theorem*)**.** *A matrix subgroup that is a regular level set has the induced structure of a Lie group. The tangent space $T_I\mathrm{SO}(3)$ is the vector space $\mathfrak{so}(3) = \{A \in \mathbb{R}^{3\times 3} : A^T = -A\}$.*

Proof. The regular level set theorem gives an embedded manifold. Matrix multiplication is polynomial in real entries. Matrix inversion is smooth on invertible matrices, by the adjugate formula divided by the nonzero determinant. These maps preserve the subgroup. In adapted submanifold charts, their restrictions retain the first manifold-coordinate components of their ambient expressions, so the restrictions are smooth. They are the group's multiplication and inversion, as required.

The orthogonal constraint and its surjective derivative were computed in Lecture 7. The determinant-one part is open in $\mathrm{O}(3)$, since its only determinants are $+1$ and -1. It therefore has the same tangent space at I. By Proposition 8.9, that space is the kernel of $A \mapsto A^T + A$. This is precisely the antisymmetric matrices. ∎

Such a matrix has the unique form

$$A(a, b, c) = \begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix}, \qquad (a, b, c) \in \mathbb{R}^3.$$

Thus $\mathfrak{so}(3)$ has basis $A(1, 0, 0)$, $A(0, 1, 0)$ and $A(0, 0, 1)$. These are infinitesimal rotations in the fixed Cartesian basis.

Definition 8.17 (Lie algebra)**.** A *Lie algebra* is a real vector space with an antisymmetric bilinear bracket satisfying the Jacobi identity $[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$.

For a Lie group embedded as a matrix subgroup, the tangent space $\mathfrak{g} = T_I G$ carries the matrix commutator $[A, B] = AB - BA$. To show closure, choose curves $a(s)$ and $b(t)$ in the group with initial value I and velocities A and B. Differentiating $a(s)b(t)a(s)^{-1}$ first in t gives $a(s)Ba(s)^{-1} \in \mathfrak{g}$. Differentiating in s at zero gives $AB - BA$, since $(a^{-1})'(0) = -A$, obtained by

differentiating $a(s)a(s)^{-1} = I$. This derivative stays in the finite-dimensional vector subspace $\mathfrak{g}$. Bilinearity and antisymmetry follow from the formula. One nested commutator expands as

$$[A, [B, C]] = ABC - ACB - BCA + CBA.$$

In the sum with its two cyclic versions, each product occurs once with each sign. This proves Jacobi. For an abstract Lie group, Lecture 9 constructs the bracket on T_eG using vector fields.

Example 8.18 (The commutator in $\mathfrak{so}(3)$)**.** We compute the bracket of the chosen rotation generators. For a vector $u = (a, b, c)$ and $w \in \mathbb{R}^3$, matrix multiplication gives

$$A(a, b, c)w = u \times w.$$

Thus this matrix represents the cross-product map A_u of Lecture 2. For a unit axis n, differentiating Lecture 2's rotation $R(tn)$ at $t = 0$ gives $A(n)$. This identifies the matrix with an initial rotation velocity. Write $E_1 = A(1, 0, 0)$, $E_2 = A(0, 1, 0)$ and $E_3 = A(0, 0, 1)$. Their products are

$$E_1E_2 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad E_2E_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Subtracting gives $[E_1, E_2] = E_3$. The same multiplication with the cyclic index choices gives $[E_2, E_3] = E_1$ and $[E_3, E_1] = E_2$. Bilinearity now gives the general formula

$$[A(u), A(w)] = A(u \times w), \qquad u, w \in \mathbb{R}^3.$$

The chosen basis therefore identifies this Lie algebra with $\mathbb{R}^3$ equipped with the cross product.

8.6 Actions and homogeneous spaces

Rotations send the north pole $N = e_3$ to every point of S^2. A rotation describing a point is not unique. If $gN = p$ and $hN = N$, then $(gh)N = gN = p$. Thus rotations that differ on the right by a rotation fixing N describe the same point. This gives a way to describe sphere points using rotations and their redundancy. Lecture 2 defined group actions on spaces. We now require the action map to be smooth. Write $R_y(t)$ and $R_z(t)$ for right-handed rotations by t about the respective Cartesian axes. The figure shows the equal images and the corresponding change of frame.

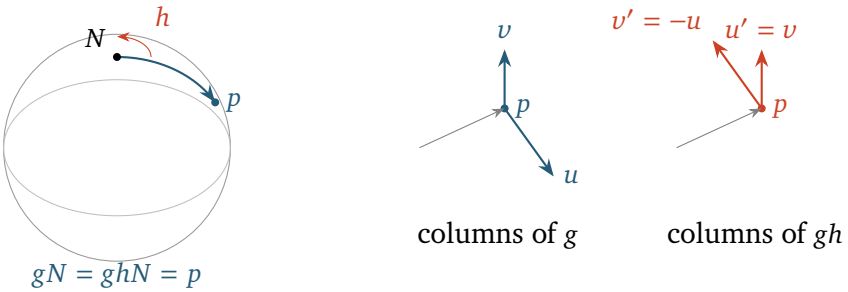


Figure 8.5: For $g = R_y(\pi/3)$ and $h = R_z(\pi/2)$, both g and gh send N to $p = (\sqrt{3}/2, 0, 1/2)$. The first two columns change from (u, v) to $(v, -u)$ while the last column stays fixed.

Definition 8.19 (Smooth left action)**.** A *smooth left action* of a Lie group on M is a smooth map $G \times M \to M$, written $(g, p) \mapsto g \cdot p$, satisfying $e \cdot p = p$ and $(gh) \cdot p = g \cdot (h \cdot p)$.

Definition 8.20 (Orbit and orbit map)**.** The *orbit* of p is $\{g \cdot p : g \in G\}$. The *orbit map* at p is the map $G \to M$ given by $g \mapsto g \cdot p$.

Definition 8.21 (Homogeneous space)**.** The action is *transitive* when the orbit of some point $p \in M$ is all of M. A manifold with a transitive smooth action is a *homogeneous space*.

For the rotation action, every unit vector is the image of N. Thus S^2 is homogeneous under $\mathrm{SO}(3)$.

Definition 8.22 (Stabiliser)**.** The *stabiliser* (or isotropy subgroup) of p is

$$H = \{g \in G : g \cdot p = p\}.$$

It is a subgroup by the action laws. It is closed, being the inverse image of the closed singleton $\{p\}$ under the continuous orbit map.

Definition 8.23 (Cosets and quotient)**.** For a subgroup H, the *left coset* gH is $\{gh : h \in H\}$. The set of left cosets, with the quotient topology, is written G/H.

Two elements have the same coset exactly when $g_1^{-1} g_2 \in H$.

Definition 8.24 (Embedded Lie subgroup)**.** An *embedded Lie subgroup* is a subgroup with an embedded smooth manifold structure and the restricted smooth group operations.

Theorem 8.25 (Closed subgroups and homogeneous spaces · *cited*)**.** *A closed subgroup H of a Lie group is an embedded Lie subgroup. The quotient G/H has a smooth manifold structure of dimension $\dim G - \dim H$ for which the quotient map is a smooth submersion. For a transitive smooth action on M with stabiliser H at p, the map $gH \mapsto g \cdot p$ is a diffeomorphism $G/H \to M$.*

These are Lee, second edition, Theorems 20.12, 21.17 and 21.18. The action identifies cosets with points: if $g_1 \cdot p = g_2 \cdot p$, then $g_1^{-1} g_2 \in H$, and the converse follows from the action law. Transitivity gives surjectivity, completing the bijection. The smooth-manifold and diffeomorphism assertions require the cited theorems. The quotient description supplies a smooth structure and a dimension formula for a space described by symmetries.

Remark 8.26 (A nonclosed subgroup)**.** Closedness cannot be omitted. A quotient by a dense proper subgroup is not Hausdorff, as in Lecture 2's quotient $\mathbb{R}/\mathbb{Q}$. For the irrational line of Lecture 7, the same obstruction occurs in a torus. Write that line as

$$H = \{(e^{it}, e^{i\alpha t}) : t \in \mathbb{R}\} \subset T^2, \qquad \alpha \in \mathbb{R} \setminus \mathbb{Q}.$$

Adding the parameters multiplies the circle components, so H is a subgroup. We prove that it is dense and proper.

The pigeonhole argument of Lecture 7 gives nonzero integer multiples of α arbitrarily close to integers. Irrationality makes each distance positive. By changing the integer's sign when necessary, we obtain a fractional part δ with $0 < \delta < \varepsilon$, for any $\varepsilon > 0$. The numbers $0, \delta, 2\delta, \ldots$, up to the last below 1, approximate every number in $[0, 1)$ to within ε. Each is

the fractional part of an integer multiple of α. Thus the points $e^{2\pi i\alpha m}$, with $m \in \mathbb{Z}$, are dense in the circle.

Now fix a target $(e^{ia}, e^{ib}) \in T^2$. The parameters $t = a + 2\pi m$ give points of H with coordinates

$$(e^{ia},\ e^{i\alpha a}e^{2\pi i\alpha m}), \qquad m \in \mathbb{Z}.$$

Their first coordinate is exactly the target's first coordinate. The density just proved lets their second coordinate approach e^{ib}. This proves density in the torus. Over first coordinate 1, however, H contains only the countably many second coordinates $e^{2\pi i\alpha m}$. The circle is uncountable, so H is proper. It is therefore not closed.

If T^2/H were Hausdorff, its singleton identity coset would be closed. Its inverse image H under the quotient map would then be closed, a contradiction. This quotient is not a manifold of our definition.

8.7 The sphere and the Hopf map

The standard action of $\mathrm{SO}(n+1)$ on S^n is transitive. Given a unit vector, extend it to an oriented orthonormal basis by Gram–Schmidt, recalled from *Linear Algebra for Physics*. Use it as the last column of an orthogonal matrix, completing the other columns with positive orientation. That matrix sends the last coordinate unit vector to the chosen point. The stabiliser consists of matrices with a block in $\mathrm{SO}(n)$ on its orthogonal complement and a last diagonal entry 1. The group $\mathrm{O}(n+1)$ is a regular level set by Lecture 7, so it is a Lie group by Proposition 8.16. Its determinant-one part is open and hence is also a Lie group. The sphere action is the restriction of the polynomial map $(g, p) \mapsto gp$, so it is smooth. Applying the cited theorem gives

$$S^n \cong \mathrm{SO}(n+1)/\mathrm{SO}(n) \qquad (n \geq 1).$$

The identification is a diffeomorphism. In particular, the orbit map $g \mapsto ge_3$ exhibits $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)$. The subgroup $\mathrm{SO}(2)$ rotates the other two columns while leaving the last one fixed. These quotient descriptions concern smooth manifolds; the stabiliser is not a normal subgroup in general, so the quotient need not be a group.

Lecture 4 identified $\mathrm{SU}(2)$ with S^3 using real coordinates. If we call its real scalar a_4, its first column is $(a_4 - iv_3, v_2 - iv_1)^T$. We now use the complex entries a and b of that column as coordinates. This gives the same identification with S^3.

The Lie group. If $|a|^2 + |b|^2 = 1$, the matrix with that first column is

$$U(a,b) = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}.$$

Its columns are orthonormal and its determinant is one, so $U(a,b) \in \mathrm{SU}(2)$. Conversely, a unitary matrix with first column $(a,b)^T$ has second column $\lambda(-\bar{b}, \bar{a})^T$ for some $|\lambda| = 1$. Its determinant is λ, so determinant one forces $\lambda = 1$. Thus every matrix in $\mathrm{SU}(2)$ has this form. Transporting the smooth structure of S^3 through U makes $\mathrm{SU}(2)$ a smooth manifold. Its product and inverse are smooth: the first columns of the corresponding matrix formulas are polynomial in the real and imaginary coordinates. This makes $\mathrm{SU}(2)$ a Lie group.

The action. We use the left action $g \cdot [a : b] = [g(a,b)^T]$ of $\mathrm{SU}(2)$ on $\mathbb{C}\mathrm{P}^1$. The notation means the line spanned by the image column. It is independent of the nonzero scalar used to represent $[a : b]$. The action is smooth in projective charts: each output ratio is a ratio of complex linear expressions, on the open set where its denominator is nonzero. Every point $[a : b]$ has a unit representative $(a,b)^T$. The matrix $U(a,b)$ sends $[1 : 0]$ to $[a : b]$, so the action is transitive.

The stabiliser. Fixing $[1 : 0]$ requires the first column to be $(e^{it}, 0)^T$. Unitarity and determinant one then force the other diagonal entry. Thus the stabiliser is the subgroup

$$H = \left\{ \begin{pmatrix} e^{it} & 0 \\ 0 & e^{-it} \end{pmatrix} : t \in \mathbb{R} \right\} \cong \mathrm{U}(1).$$

Consequently the cited theorem applies and gives $\mathbb{C}\mathrm{P}^1 \cong \mathrm{SU}(2)/\mathrm{U}(1)$ as smooth manifolds.

The figure places the two orbit maps and their stabilisers side by side.

$$\begin{array}{ccc} \mathrm{SO}(2) & \longrightarrow & \mathrm{SO}(3) \\ & & \downarrow{\scriptstyle g \mapsto ge_3} \\ & & S^2 \end{array} \qquad \begin{array}{ccc} \mathrm{U}(1) & \longrightarrow & \mathrm{SU}(2) \\ & & \downarrow{\scriptstyle g \mapsto [ge_1]} \\ & & \mathbb{C}\mathrm{P}^1 \end{array}$$

Figure 8.6: Each orbit map retains one datum: the last unit column for a rotation, or the line of the first unit column for $\mathrm{SU}(2)$. Multiplication on the right by its stabiliser leaves that datum fixed.

Definition 8.27 (Hopf map). The *Hopf map* sends a unit column to its complex line:

$$S^3 \longrightarrow \mathbb{C}\mathrm{P}^1, \qquad (a,b) \longmapsto [a : b].$$

The two orbit maps have circle fibres. For a fixed image $n = ge_3$, the rotations are $gR_z(t)$ with t taken modulo 2π. Locally, choosing a reference rotation for each nearby direction lets us describe a rotation by that direction and a spin angle about it. For a fixed projective point $[a : b]$, the unit columns are $e^{it}(a,b)^T$ with the same period. The figure draws these parameter circles over two chosen base points.

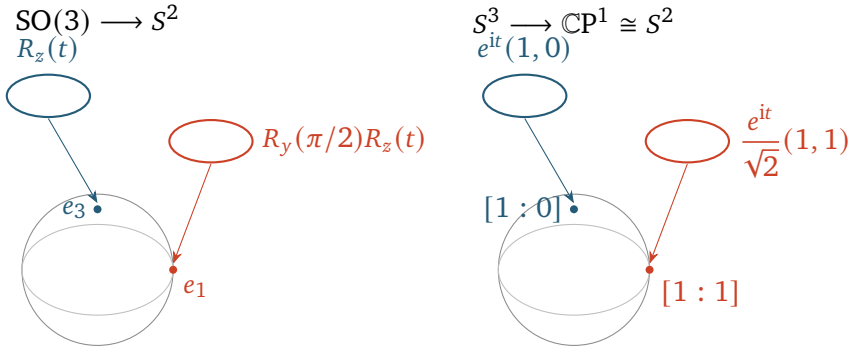


Figure 8.7: The displayed circles are schematic copies of the individual fibres: $g\mathrm{SO}(2)$ for rotations and $\{e^{it}(a,b)^T\}$ for the Hopf map. Their placement above the spheres does not assert a global product.

Example 8.28 (The Hopf map · *proved given the homogeneous-space theorem*)**.** The first-column map identifies SU(2) with $S^3 = \{(a,b) \in \mathbb{C}^2 : |a|^2 + |b|^2 = 1\}$. The inverse is $U(a,b)$, so both maps are smooth. The projection to the first-column line is the Hopf map. Its fibre consists of the unit representatives $(e^{it}a, e^{it}b)$ of the same complex line. Indeed, two unit columns span the same line exactly when the scalar relating them has modulus one. The matrix identity $U(e^{it}a, e^{it}b) = U(a,b)\,\mathrm{diag}(e^{it}, e^{-it})$ shows that these are precisely the left cosets $U(a,b)H$, the fibres of the orbit map $g \mapsto [ge_1]$. Composing with the Bloch diffeomorphism of Example 7.15 gives $S^3 \to S^2$. It forgets the common phase of a normalised qubit column. Lecture 20 develops its structure as a circle bundle and proves that no continuous choice of a normalised representative exists over all projective states.

Self-check. Find the dimension of $\mathrm{SO}(3)/\mathrm{SO}(2)$ using its stabiliser, and compare it with the tangent-plane calculation for the sphere. Then let SO(3) act on $\mathbb{R}\mathrm{P}^2$ by $g \cdot [X] = [gX]$. Find the stabiliser of $[0:0:1]$, its connected components and the quotient dimension.

Answer. The matrices in $\mathfrak{so}(3)$ have three independent parameters. The group SO(2) is a circle, so it has dimension one. The quotient dimension is $3 - 1 = 2$. The plane $p \cdot v = 0$ in $\mathbb{R}^3$ also has dimension two, since $p \neq 0$. The two calculations agree.

The projective action is transitive because the sphere action is transitive. Fixing $[0:0:1]$ means $ge_3 = \pm e_3$. Orthogonality preserves the perpendicular plane. Hence the stabiliser is

$$\left\{ \begin{pmatrix} B & 0 \\ 0 & \det B \end{pmatrix} : B \in \mathrm{O}(2) \right\}.$$

Indeed, the determinant-one condition forces the last entry to be $\det B$. The two components have $\det B = 1$ and $\det B = -1$. The first consists of rotations about e_3. The second consists of half-turns about axes in its perpendicular plane. Both have dimension one, so the quotient again has dimension $3 - 1 = 2$.

Summary. Tangent vectors are velocity classes. A chart identifies them with coordinate columns, and its Jacobian gives their transformation law. Differentials send curve velocities to image velocities and satisfy the chain rule. The tangent space of a regular level set is the kernel of the constraint's derivative. Closed stabilisers give smooth quotient manifolds by cited theorems. The rotation and qubit actions identify the sphere with $\mathrm{SO}(3)/\mathrm{SO}(2)$ and $\mathrm{SU}(2)/\mathrm{U}(1)$ respectively. Lecture 9 studies curves whose velocities are prescribed at every point by a vector field.

Reading. Tu, *An Introduction to Manifolds*, §§8, 12, 15–16; Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 20.12, 21.17 and 21.18.